

Assignment-1

MVC Design pattern and its Variants

Overview

The Model-view-controller (MVC) design pattern divides an application into three interconnected components:

- Model: - Manages the data and business logic.
- View: - Handles the presentation layer and user interface.
- Controller: - processes user input and interacts with the Model and View.

This separation facilitates modular development, easier testing and maintenance.

Variants of MVC

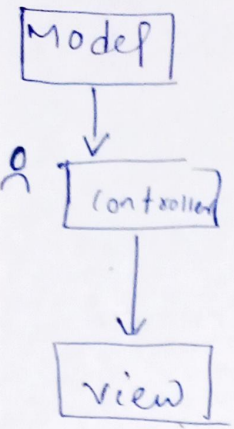
1. Model-view-presenter (MVP):

- Presenter: Acts as an intermediary between the view and Model, handling all presentation logic.
- use case: suitable for applications requiring clear separation between UI and business logic such as windows forms application.

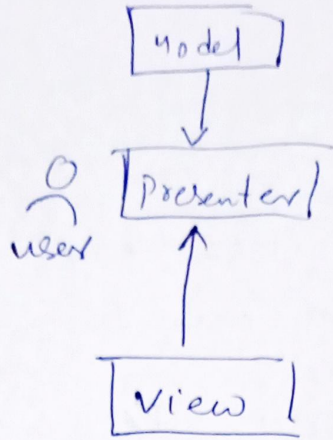
2. Model-view-viewmodel (MVVM):

- viewModel: Exposes data and commands from the Model to the view, facilitating data binding.
- use case: ideal for applications using data-binding capabilities, like WPF or Angular application.

MVC



MVP



MVVM

